Versal Deformation Algorithm for the First Companion Form

MATLAB Implementation with Orthonormalization
(Updated according to ALP 2026 and SIAM 2003)

Abstract

This document summarizes all mathematical formulas implemented in the MATLAB code for constructing versal deformations of Fiedler linearizations (first companion form). The implementation incorporates a **numerically robust orthonormalization** of the bases S_t and $\mathcal{V}_s$ via singular value decomposition (SVD), which guarantees that the recurrence matrix K becomes the identity and improves conditioning. All references correspond to the equations in the article ALP 2026 and the classical work by García-Planas and Mailybaev (SIAM 2003). The complete code is available as supplementary material.

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1 Companion Matrices of the Polynomial

For a quadratic scalar polynomial $P(\lambda) = a_2\lambda^2 + a_1\lambda + a_0$, the companion matrices are (Eq. (1) in ALP 2026):

$$A_0 = \begin{bmatrix} a_1 & a_0 \\ -1 & 0 \end{bmatrix} \in \mathbb{R}^{2\times 2} \tag{1}$$

$$B_0 = \begin{bmatrix} -a_2 & 0 \\ 0 & -1 \end{bmatrix} \in \mathbb{R}^{2\times 2} \tag{2}$$

MATLAB function: `step0_build_Ao_Bo.m`

2 Tangent Matrix T

The matrix T represents the differential df_{X_0} of the orbit tangent space (Eq. (11) in ALP 2026):

$$T = \begin{bmatrix} A_0^\mathsf{T} \otimes I_p & -I_q \otimes A_0 \\ B_0^\mathsf{T} \otimes I_p & -I_q \otimes B_0 \end{bmatrix} \in \mathbb{R}^{2pq\times(p^2+q^2)} \tag{3}$$

For the scalar quadratic case, $p = q = 2$, so $T \in \mathbb{R}^{8\times 8}$.
MATLAB function: `step1_build_T.m`

3 Constraint Matrix T_1

The matrix T_1 encodes the conditions for the optimal subspace S_t (Corollary 3.1 and Eq. (14) in ALP 2026):

$$T_1 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & -1 & 0 & 0 & -1 \end{bmatrix} \in \mathbb{R}^{3\times 8} \tag{4}$$

Interpretation of rows:

- Row 1: $v_{21} = 0$ (last block row of V is zero)
- Row 2: $v_{22} = 0$ (last block row of V is zero)
- Row 3: $\mathrm{tr}_2(U) = \mathrm{tr}_2(V)$ (trace condition)

MATLAB function: `step2_build_T1.m`

4 Robust Construction of S_t and $\mathcal{V}_s$ (Orthonormalization)

To ensure numerical stability and $K = I$, the SVD-based procedure described in Section 5 ("Numerical stabilization of the bases") is used:

1. Compute an orthonormal basis of $\ker T_1$:

$$Z = \mathrm{orth}(\ker T_1) \in \mathbb{R}^{8\times 5} \tag{5}$$

2. Apply the tangent matrix: $A = TZ \in \mathbb{R}^{8\times 5}$

3. SVD of A:

$$A = U\Sigma V^{\mathsf{T}} \tag{6}$$

4. The orthonormal bases are:

$$S_t = U \qquad \text{(orthonormal columns)} \tag{7}$$
$$\mathcal{V}_s = ZV\Sigma^{-1} \qquad \text{(such that } T\mathcal{V}_s = S_t\text{)} \tag{8}$$

Properties:

- $S_t^{\mathsf{T}} S_t = I$ (orthonormal)
- $T\mathcal{V}_s = S_t$
- $K = S_t^{\mathsf{T}}(T\mathcal{V}_s) = I$ (identity)

Implementation: main script (steps 4 and 6) in `versal_deformation_main.m`

5 Complementary Subspaces

5.1 Orthogonal complement $S_t^{\perp}$

$$S_t^{\perp} = \mathrm{null}(S_t^{\mathsf{T}}) \quad \text{(orthogonal within } T_{X_0}\mathcal{O}(X_0)\text{)} \tag{9}$$

Dimension: $8 - 5 = 3$ basis vectors.

5.2 Orthogonal complement $\mathcal{V}_s^{\perp}$

$$\mathcal{V}_s^{\perp} = \mathrm{null}(\mathcal{V}_s^{\mathsf{T}}) \quad \text{(orthogonal within } T_e\mathcal{G}\text{)} \tag{10}$$

Dimension: $8 - 5 = 3$ basis vectors.

MATLAB function: `step5_orthogonal_complement.m`

6 Generalized Sylvester Space $\mathcal{GS}$

Basis of the space of structured perturbations (affecting only polynomial coefficients), see Appendix A in ALP 2026:

$$\mathcal{GS} = \mathrm{span}\left\{ \begin{bmatrix}1\\0\\0\\0\\0\\0\\0\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\\0\\0\\0\\0\\0\end{bmatrix}, \begin{bmatrix}0\\0\\0\\0\\-1\\0\\0\\0\end{bmatrix} \right\} \tag{11}$$

Correspondence:

- Column 1: perturbation of F_1 (a_1) in the A block
- Column 2: perturbation of F_0 (a_0) in the A block
- Column 3: perturbation of F_2 (a_2) in the B block (negative sign due to the definition of B_0)

MATLAB function: `step7_generalized_Sylvester.m`

7 Recurrence Matrices L and K

7.1 Matrix L (3×3)

$$L_{ij} = \langle C_j, N_i \rangle_1 = C_j^{\mathsf{T}} N_i \tag{12}$$

where $\{C_j\}$ is a basis of $\mathcal{GS}$ and $\{N_i\}$ a basis of $S_t^{\perp}$. (Eq. (29) in ALP 2026)

7.2 Matrix K (5×5)

$$K_{ij} = \langle df_{X_0}(R_j), \mathcal{T}_i \rangle_1 \tag{13}$$

with $\{R_j\}$ basis of $\mathcal{V}_s$, $\{\mathcal{T}_i\}$ basis of S_t. (Eq. (30) in ALP 2026)

The computation of $df_{X_0}(R_j)$ for $R_j = U_j - \lambda V_j$ uses Eq. (9):

$$df_{X_0}(R_j) = \begin{bmatrix} A_0 V_j - U_j A_0 \\ B_0 V_j - U_j B_0 \end{bmatrix} \tag{14}$$

Important property: Due to the orthonormal construction, $K = I$ (identity).
MATLAB function: `step8_build_L_K.m`

8 First-Order Derivatives

8.1 Derivatives of φ (polynomial parameters)

Using Eq. (31) in ALP 2026:

$$\begin{bmatrix} D^{\mathbf{e}_k}\varphi_0 \\ D^{\mathbf{e}_k}\varphi_1 \\ D^{\mathbf{e}_k}\varphi_2 \end{bmatrix} = L^{-1} \begin{bmatrix} \langle \mathbf{e}_k, N_1 \rangle_1 \\ \langle \mathbf{e}_k, N_2 \rangle_1 \\ \langle \mathbf{e}_k, N_3 \rangle_1 \end{bmatrix} = L^{-1} N^{\mathsf{T}} \mathbf{e}_k \tag{15}$$

for $k = 1, \ldots, 8$ (perturbation parameters).

8.2 Derivatives of μ (transformation coefficients)

Using Eq. (32) in ALP 2026:

$$\begin{bmatrix} D^{\mathbf{e}_k}\mu_1 \\ \vdots \\ D^{\mathbf{e}_k}\mu_5 \end{bmatrix} = K^{-1} \begin{bmatrix} \langle \mathbf{e}_k - \sum_{j=0}^{2} D^{\mathbf{e}_k}\varphi_j C_j,\ \mathcal{T}_1 \rangle_1 \\ \vdots \\ \langle \mathbf{e}_k - \sum_{j=0}^{2} D^{\mathbf{e}_k}\varphi_j C_j,\ \mathcal{T}_5 \rangle_1 \end{bmatrix} \tag{16}$$

MATLAB function: `step9_derivatives.m` (first loop)

9 Three-Argument Function α

The α function is defined in Eq. (34) of ALP 2026 (and Eq. (2.23) of SIAM 2003):

$$\alpha(X_1, Y_1, Z_1) = (AV - UX) - \lambda(BV - UY) \tag{17}$$

where:

- $X_1 = A - \lambda B$ belongs to $\mathcal{GS}$ (Sylvester space)
- $Y_1 = (U, V)$ belongs to $T_e\mathcal{G}$ (space $\mathcal{V}_s$)
- $Z_1 = X - \lambda Y$ is the perturbation direction

MATLAB implementation: `galfa_three_correcto.m`

Listing 1: galfa_three_correcto.m

```
function val = galfa_three_correcto(X, Y, Z, Ao,
    Bo)
% X: 8x1 vector (A_X, B_X) from GS
% Y: 8x1 vector (U, V) from Vs
% Z: 8x1 vector (X_Z, Y_Z) perturbation direction
m = 2; n = 2;
A_X = reshape(X(1:m*n), m, n);
B_X = reshape(X(m*n+1:end), m, n);
U = reshape(Y(1:m*m), m, m);
V = reshape(Y(m*m+1:end), n, n);
X_Z = reshape(Z(1:m*n), m, n);
Y_Z = reshape(Z(m*n+1:end), m, n);
DeltaA = A_X * V - U * X_Z;
DeltaB = B_X * V - U * Y_Z;
val = [DeltaA(:); DeltaB(:)];
end
```

10 Second-Order Derivatives

10.1 Inhomogeneous term S_h

Using Eq. (33) in ALP 2026 (and Eq. (2.22) in SIAM 2003):

$$S_h = D^h(X - \lambda Y) - \sum_{\substack{h_1+h_2=h \\ h_1,h_2\neq h}} \binom{h}{h_1} \alpha\left(\sum_i D^{h_1}\varphi_i C_i,\ \sum_j D^{h_2}\mu_j R_j,\ D^{h_1}(X - \lambda Y)\right) \tag{18}$$

10.2 Binomial coefficients for second order ($|h| = 2$)

- If $h = \mathbf{e}_i + \mathbf{e}_j$ with $i \neq j$: two terms each with coefficient $\binom{2}{1} = 2$.
- If $h = 2\mathbf{e}_i$: one term with coefficient $\binom{2}{1} = 2$.

10.3 Second derivatives of φ

Using Eq. (31) applied to $|h| = 2$:

$$\begin{bmatrix} D^h\varphi_0 \\ D^h\varphi_1 \\ D^h\varphi_2 \end{bmatrix} = L^{-1} \begin{bmatrix} \langle S_h, N_1\rangle_1 \\ \langle S_h, N_2\rangle_1 \\ \langle S_h, N_3\rangle_1 \end{bmatrix} \tag{19}$$

10.4 Second derivatives of μ

Using Eq. (32) applied to $|h| = 2$:

$$\begin{bmatrix} D^h\mu_1 \\ \vdots \\ D^h\mu_5 \end{bmatrix} = K^{-1} \begin{bmatrix} \langle S_h - \sum_{k=0}^{2} D^h\varphi_k C_k,\ \mathcal{T}_1\rangle_1 \\ \vdots \\ \langle S_h - \sum_{k=0}^{2} D^h\varphi_k C_k,\ \mathcal{T}_5\rangle_1 \end{bmatrix} \tag{20}$$

The second derivatives are stored in tensors $B^k \in \mathbb{R}^{8\times 8}$ (for φ_k) and $B^{\mu k} \in \mathbb{R}^{8\times 8}$ (for μ_k).

MATLAB function: `step9_derivatives.m` (second part)

11 Supplementary Material

The complete MATLAB code is provided as supplementary material. It includes:

- Functions for each step of the algorithm.
- A main script that automatically generates LaTeX files with the orthonormal bases.
- An additional script that produces a comparison between raw (non-orthonormal) and orthonormal bases.
- Numerical verification that $K = I$ (identity) after orthonormalization.

Generated files:

- `versal_deformation_output.tex` – contains the orthonormal matrices ready for inclusion in the article.
- `bases_comparison_supplement.tex` – shows both raw and normalized versions for transparency.

12 Numerical Verification (Example $a_0 = 1, a_1 = 2, a_2 = 3$)

Concept	**Dimension / Value**	**Observation**

T (Eq. (11))	8×8, full rank	$\mathrm{rank}(T) = 8$
T_1 (Eq. (14))	3×8, $\mathrm{rank}(T_1) = 3$	
$\ker T_1$ (Lemma 3.1)	8×5	5 basis vectors
S_t (orthonormal)	8×5, $S_t^\mathsf{T} S_t = I$	$K = I$
$S_t^\perp$	8×3	orthogonal complement
$\mathcal{V}_s$ (orthonormal)	8×5, $T\mathcal{V}_s = S_t$	constructed via SVD
$\mathcal{GS}$ (Appendix A)	8×3	3 basis vectors
L (Eq. (29))	3×3, invertible	$\det(L) \approx -0.007$
K (Eq. (30))	5×5	**Identity** (error $< 10^{-14}$)

First-order Taylor coefficients (Eq. (31) and (32)):

$$\varphi_0(\vec{\epsilon}) = 0.955\epsilon_1 + 0.126\epsilon_2 + 0.063\epsilon_3 + 0.502\epsilon_4 - 0.230\epsilon_5 + 0.063\epsilon_8 + \dots$$
$$\varphi_1(\vec{\epsilon}) = 0.362\epsilon_1 - 1.774\epsilon_2 - 0.887\epsilon_3 - 3.393\epsilon_4 + 2.018\epsilon_5 - 0.887\epsilon_8 + \dots$$
$$\varphi_2(\vec{\epsilon}) = 0.250\epsilon_1 + 0.802\epsilon_2 + 0.401\epsilon_3 - 1.804\epsilon_4 + 0.200\epsilon_5 + 0.401\epsilon_8 + \dots$$

Quadratic terms are incorporated in the final series presented in the article (Section 7, Step 10).

13 Cross-References Between Code and Article

MATLAB function	Equation in ALP 2026	Description
`step0_build_Ao_Bo`	(1)	Companion matrices
`step1_build_T`	(11)	Tangent matrix T
`step2_build_T1`	Corollary 3.1 + (14)	Constraint matrix T_1
`step3_kernel_T1`	Lemma 3.1	$\ker T_1$
`(main script)`	Section 5 (stabilization)	SVD construction of S_t, $\mathcal{V}_s$
`step5_orthogonal_complement`	(20)	$S_t^\perp$, $\mathcal{V}_s^\perp$
`step7_generalized_Sylvester`	Appendix A	Basis of $\mathcal{GS}$
`step8_build_L_K`	(29)–(30)	Matrices L, K
`step9_derivatives`	(31)–(32), (33)–(34)	First and second derivatives
`galfa_three_correcto`	(34) ALP, (2.23) SIAM	α with 3 arguments

14 Conclusion

The presented implementation:

- Faithfully follows the equations of ALP 2026 (Eq. (1), (11), (14), (29)–(30), (31)–(34), etc.) and the classical work by García-Planas and Mailybaev (SIAM 2003).
- Incorporates a **numerically robust orthonormalization** via SVD (Section 5 of ALP 2026), guaranteeing $K = I$ and excellent conditioning.
- Provides complete source code as supplementary material, including automatic generation of LaTeX tables.

- Has been verified with the scalar quadratic example, reproducing exactly the coefficients published in the article.

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